Peak vs. Non-Peak Performance of Commercial LLM APIs: A 28-Day Measurement Study

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May 2026

Abstract

Commercial large language model (LLM) APIs are shared infrastructure, yet little is known about how server load during peak business hours affects latency, throughput, and answer quality in practice. We benchmark four providers—OpenAI (`gpt-4o-mini`), Anthropic (`claude-haiku-4-5`), Google (`gemini-flash-lite`), and Alibaba Qwen (`qwen3.5-flash`)—twice daily for 28 days using 30 MMLU questions, measuring time-to-first-token (TTFT), tokens-per-second (TPS), and answer accuracy. Peak sessions run at 10:00 ET (within the 8 AM–2 PM US East Coast peak band); non-peak sessions run at 22:00 ET. We find that TTFT increases significantly during peak hours for US-hosted providers (+4% to +37%), while Qwen—served from an Alibaba Cloud US endpoint—is counter-intuitively *faster* during US peak hours (−7% TTFT, +9% TPS), consistent with the hypothesis that US peak corresponds to Asia off-peak. Answer accuracy shows no statistically significant change for any provider across conditions, suggesting that output quality is decoupled from infrastructure load at the scale tested.

1 Introduction

Large language model APIs have become commodity infrastructure for production applications, yet their service-level properties under varying load conditions remain largely undocumented from the consumer's perspective. Unlike traditional web services whose latency characteristics are well-studied, LLM inference involves substantial per-request compute that amplifies the effect of concurrent demand.

Traffic on commercial AI APIs follows predictable diurnal patterns driven by North American business hours. If providers do not over-provision sufficiently, practitioners who call APIs during peak hours may receive degraded latency—increasing costs for time-sensitive applications such as real-time agents or interactive chat. Whether throughput degradation also affects output *quality* (answer accuracy) is an open question with direct implications for reliability-critical deployments.

This paper reports the results of a 28-day empirical study comparing peak-hour and off-peak-hour performance across four major LLM API providers on a standardized MMLU benchmark suite. Our primary contributions are:

1. A reproducible measurement framework for longitudinal LLM API benchmarking.
2. Provider-specific quantification of peak-vs-nonpeak latency and throughput effects.
3. Evidence that MMLU accuracy is robust to infrastructure load conditions.

4. An anomalous finding for a non-US-headquartered provider suggesting inter-regional load balancing.

2 Methodology

2.1 Providers and Models

We evaluated four providers and their smallest fast-inference models available at the time of the study (Table 1). All models were prompted with a fixed system prompt instructing single-letter responses (`A`–`D`), with `temperature=0` for reproducibility. Streaming inference was used for all calls to measure TTFT directly.

Table 1: Models under evaluation.

Provider	Model	Endpoint region
OpenAI	`gpt-4o-mini-2024-07-18`	US East
Anthropic	`claude-haiku-4-5-20251001`	US East
Google	`gemini-3.1-flash-lite-preview`	US (GCP)
Qwen	`qwen3.5-flash`	US (Alibaba Cloud)

2.2 Question Set

We selected 30 questions from the Massive Multitask Language Understanding (MMLU) benchmark [Hendrycks et al., 2021], sampling across subjects and difficulty levels. Each question was repeated three times per session to reduce variance, yielding 90 trials per provider per session (30 questions $\times$ 3 repeats). Accuracy was scored by extracting the first valid option letter from the model response using a cascade of regular expressions targeting XML-tagged answers, explicit "Answer:" labels, and trailing isolated letters.

2.3 Experimental Schedule

Experiments ran from 2026-03-30 to 2026-04-26 (28 days). Two sessions were scheduled daily in the America/New_York timezone:

- **Peak**: 10:00 ET — within the 8 AM–2 PM Eastern peak band.
- **Non-peak**: 22:00 ET — US East Coast late night.

Sessions were managed by a Python scheduler (APScheduler) running as a launchd daemon on a Mac Mini with a residential internet connection. Progress was checkpointed per task to support resumption after interruptions.

2.4 Metrics

- **TTFT** (time-to-first-token, seconds): wall-clock time from request dispatch to receipt of the first streamed token.
- **TPS** (tokens per second): output token count divided by generation time after the first token.
- **Accuracy**: proportion of trials where the parsed answer matches the correct MMLU key.

2.5 Statistical Analysis

All pairwise comparisons use the Mann–Whitney U test (two-sided), which makes no distributional assumptions and is robust to the heavy-tailed latency distributions observed. Effect sizes are reported as median differences and percent change relative to the non-peak median (mean for accuracy). Significance threshold: $\alpha = 0.05$.

3 Results

3.1 Data Completeness

Of 56 scheduled slots (28 days $\times$ 2 periods), 54 were executed; 3 non-peak slots were missed due to daemon downtime. Most executed slots were partial: API errors (primarily Google returning HTTP 429/503) caused premature termination in 45 of 54 slots. A *complete* slot is defined as exactly 360 rows (30 questions $\times$ 3 repeats $\times$ 4 providers). Only 7 slots were complete (3 non-peak, 4 peak). All primary results below use these 7 complete slots (1,890 rows total); see Section 5 for discussion of this constraint.

3.2 Latency: Time-to-First-Token

Table 2: TTFT comparison across conditions (complete slots only; $n_{\text{peak}} = 360$, $n_{\text{nonpeak}} = 270$ per provider; medians reported). Significance: $^{***}p < 0.001$, $^{*}p < 0.05$, ns=not significant.

Provider	Non-peak (s)	Peak (s)	Δ (s)	Δ (%)	Sig.
OpenAI	0.873	0.910	+0.037	+4.3%	*
Anthropic	0.835	0.971	+0.136	+16.3%	***
Google	1.721	2.355	+0.634	+36.8%	***
Qwen	3.698	3.445	−0.253	−6.8%	***

Three of the four providers exhibit significantly higher TTFT during peak hours (Table 2). Google shows the largest absolute and relative increase (+37%), while OpenAI shows the smallest (+4%). Qwen is the sole outlier: TTFT is *lower* during peak hours, a pattern we discuss in Section 4.

3.3 Throughput: Tokens per Second

Table 3: TPS comparison (complete slots only; medians).

Provider	Non-peak	Peak	Δ	Δ (%)	Sig.
OpenAI	54.9	53.6	−1.4	−2.5%	*ns*
Anthropic	139.5	139.1	−0.5	−0.3%	*ns*
Google	221.6	226.9	+5.3	+2.4%	*ns*
Qwen	171.3	187.3	+15.9	+9.3%	***

TPS differences are not statistically significant for the three US-hosted providers (Table 3). Qwen again shows the opposite pattern: generation throughput is significantly higher during US peak hours.

Table 4: MMLU accuracy (mean correct, complete slots only).

Provider	**Non-peak**	**Peak**	Δ	**Sig.**
OpenAI	0.493	0.475	-0.018	*ns*
Anthropic	0.833	0.817	-0.017	*ns*
Google	0.867	0.864	-0.003	*ns*
Qwen	0.822	0.808	-0.014	*ns*

3.4 Answer Accuracy

No provider shows a statistically significant change in accuracy between peak and non-peak conditions (Table 4, all $p > 0.5$). The sub-50% accuracy of OpenAI's `gpt-4o-mini` reflects this model's sensitivity to strict single-letter output formatting; format-failure rates range from 1.5% to 2.8%.

4 Discussion

US-hosted providers: TTFT scales with load. OpenAI, Anthropic, and Google all exhibit elevated TTFT during peak hours, consistent with increased queuing or scheduling overhead under higher concurrent demand. The effect ranges from modest for OpenAI ($\approx$4%) to substantial for Google ($\approx$37%), suggesting different capacity margins across providers. Critically, this latency increase does not propagate to generation speed or accuracy, implying that once the model begins generating, execution is largely isolated from request-level congestion.

Qwen anomaly: cross-regional load balancing. Qwen's US endpoint (`dashscope-us.aliyuncs.com`) runs on Alibaba Cloud infrastructure. US business peak hours (10:00 ET) correspond to approximately 23:00–01:00 the following day in China Standard Time—well outside Chinese business hours. The observed lower TTFT and higher TPS during US peak hours is consistent with the hypothesis that Alibaba's global scheduler shifts load to US-region GPUs when Asian demand is low, effectively benefiting US users during their own peak. This finding suggests that infrastructure geography and inter-regional load balancing may produce effects opposite to naive expectations for non-US-headquartered providers.

Accuracy robustness. The absence of accuracy degradation during peak hours is practically important. Applications relying on LLM APIs for classification or reasoning can expect consistent output quality regardless of the time of day. Operators optimizing purely for quality need not schedule around traffic patterns. Those optimizing for latency-sensitive workloads (real-time agents, interactive chat) should budget for TTFT increases of 4–37% during US peak hours.

5 Limitations

Limited complete-slot coverage. Of 54 executed slots, only 7 met the completeness threshold of 360 rows. Partial slots were caused predominantly by Google API rate-limiting (HTTP 429/503), which disproportionately affected non-peak slots and may introduce selection bias. Analyses are restricted to complete slots to minimize this bias, but the sample covers only a subset of the 28-day window and reduces statistical power.

Single time point per condition. Each daily session is a single snapshot (10:00 or 22:00 ET). Intra-period variability (e.g., 9:00 vs. 13:00 during peak) is not captured.

Residential network variability. All measurements were made from a Mac Mini on a residential internet connection. Network jitter contributes to TTFT variance and cannot be fully separated from server-side latency effects.

Small question set and statistical power. 30 MMLU questions are insufficient to detect small accuracy effects (<2%) or subject-level differences. A larger question set would improve sensitivity.

Model versioning. Providers may silently update model weights or infrastructure during the study period. Observed trends reflect the combined effect of load patterns and any within-study infrastructure changes, which we did not control for.

6 Conclusion

We conducted a 28-day longitudinal benchmark of four commercial LLM APIs at US business-hour peak and off-peak times. US-hosted providers (OpenAI, Anthropic, Google) show significantly elevated TTFT during peak hours (4–37%), with no corresponding degradation in generation throughput or answer accuracy. Qwen, served from a US-region Alibaba Cloud endpoint, shows the opposite pattern—faster TTFT and higher TPS during US peak—consistent with cross-regional load balancing.

These findings suggest that practitioners should expect latency increases for latency-critical applications during US business hours but can rely on consistent output quality regardless of time of day. Future work should extend the study period, increase question coverage, and instrument server-side metrics to corroborate the load-balancing hypothesis.

Data and Code Availability

The benchmark framework, collected CSVs, and analysis scripts are available at https://github.com/silverbreak1/llm_trafic_router.

References

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *Proceedings of the International Conference on Learning Representations*, 2021. URL https://arxiv.org/abs/2009.03300.